ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P1HR Worked Solutions

Jun 2025 | P1HR

33 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P1HR - Jun 2025 | P1HR

1 Find the lowest common multiple (LCM) of 45 and 70

....

(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$45 = 3^2 \times 5$$

$$70 = 2 \times 5 \times 7$$

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2 \times 3^2 \times 5 \times 7 = 630$$

FINAL ANSWER

630

PAST PAPER QUESTION Q#: 2 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P1HR - Jun 2025 | P1HR

2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

.... m

(1)

(ii) Write down the upper bound of the length of the ship.

.... m

(1)

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

The length is 142.8 m correct to 1 decimal place.
Half of 0.1 is 0.05.

$$142.8 - 0.05 = 142.75$$

$$142.8 + 0.05 = 142.85$$

FINAL ANSWER

(i) 142.75 m, (ii) 142.85 m

PAST PAPER QUESTION

Topic: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

Q#: 3 Marks: 3

3 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$2\frac{1}{4} = \frac{9}{4}, \qquad 1\frac{5}{7} = \frac{12}{7}$$

$$2\frac{1}{4} \times 1\frac{5}{7} = \frac{9}{4} \times \frac{12}{7}$$

$$= \frac{108}{28} = \frac{27}{7} = 3\frac{6}{7}$$

FINAL ANSWER

$3\frac{6}{7}$.

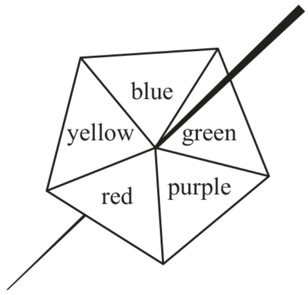
PAST PAPER QUESTION

Q#: 4 **Marks: 5**

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

4 Here is a biased 5-sided spinner.
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

Colour	blue	green	purple	red	yellow
Probability	0.12	0.20	0.38	$4x$	x

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

....
(1)

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

....
(4)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

Let the yellow probability be x. Then the red probability is $4x$.

$$0.12 + 0.20 + 0.30 + 4x + x = 1$$

$$5x = 0.38$$

$$x = 0.076$$

So

$$P(\text{red}) = 4x = 0.304$$

$$P(\text{blue, green or purple}) = 0.12 + 0.20 + 0.30 = 0.62$$

Expected red:

$$350 \times 0.304 = 106.4$$

FINAL ANSWER

0.62, and about 106.

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

5 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

....

(ii) B'

....

(2)

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

$\mathscr{E}$	$\cap$	$\cup$	$\varnothing$	$\in$	$\notin$

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C =$ ....

(ii) 13 .... $\mathscr{E}$

(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION Q#: 5 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Set notation and Venn diagrams. The tag is correct.

Method

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

Therefore

$$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$$

and

$$B' = \{1, 2, 4, 5, 7, 8, 10, 11\}$$

No number is both even and odd, so

$$A \cap C = \emptyset$$

Also $13 \notin \mathcal{E}$ because 13 is not in the universal set.

FINAL ANSWER

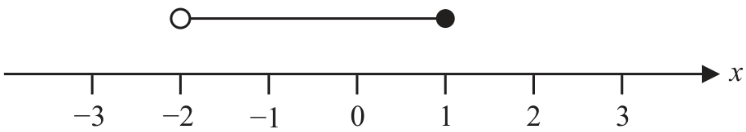
(a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$, (a)(ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$, (b)(i) $\emptyset$, (b)(ii) $13 \notin \mathcal{E}$

PAST PAPER QUESTION Q#: 6 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2025 P1HR - Jun 2025 | P1HR

6



(a) Write down the inequality shown on the number line.

....

(2)

(b) Solve the inequality $7a - 5 \leqslant 3a + 28$
Show clear algebraic working.

....

(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Solving Inequalities**

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WORKED SOLUTION

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Topic check. Solving inequalities. The tag is correct.

Solution

(a) The number line has an open circle at -2, a closed circle at 1, and shading between them.

$$-2 < x \leq 1$$

(b)

$$7a - 5 \leq 3a + 28$$

$$4a \leq 33$$

$$a \leq \frac{33}{4}$$

FINAL ANSWER

$-2 < x \leq 1$, and $a \leq \frac{33}{4}$.

PAST PAPER QUESTION

TOPIC: **Standard & Compound Units**

Jun 2025 P1HR - Jun 2025 | P1HR

Q#: 7 Marks: 3

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

.... km/h

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Correct. This is converting metres per second to kilometres per hour algebraically.

Solution

To convert m/s to km/h, multiply by 3.6.

$$50x \times 3.6 = 180x$$

FINAL ANSWER

$180x$ km/h.

PAST PAPER QUESTION

TOPIC: **Factorising**

Jun 2025 P1HR - Jun 2025 | P1HR

Q#: 8 Marks: 5

8 (a) Simplify $a^6 \times a^{10}$

....
(1)

(b) Simplify $c^{30} \div c^{12}$

....
(1)

(c) (i) Factorise $y^2 - 10y + 21$

....
(2)

(ii) Hence, solve $y^2 - 10y + 21 = 0$

....
(1)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Factorising**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

(a) $a^6 \times a^{10} = a^{16}$.
(b) $c^{30} \div c^{12} = c^{18}$.
(c)(i)

$$y^2 - 10y + 21 = (y-3)(y-7)$$

(c)(ii)

$$y = 3 \quad \text{or} \quad y = 7$$

FINAL ANSWER

a^{16}, c^{18}, $(y-3)(y-7)$, $y = 3$ or $y = 7$

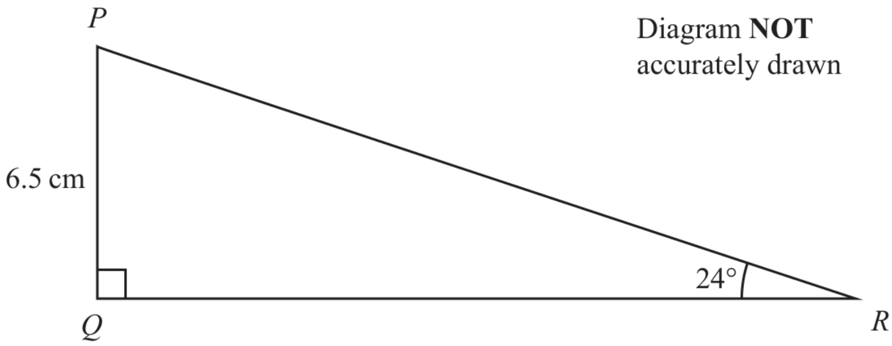
PAST PAPER QUESTION

Q#: 9 **Marks: 3**

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

9 The diagram shows triangle *PQR*



Work out the length of *QR*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry.

Solution

$$\tan 24^\circ = \frac{6.5}{QR}$$

$$QR = \frac{6.5}{\tan 24^\circ} = 14.599\ldots$$

FINAL ANSWER

14.6 cm.

PAST PAPER QUESTION

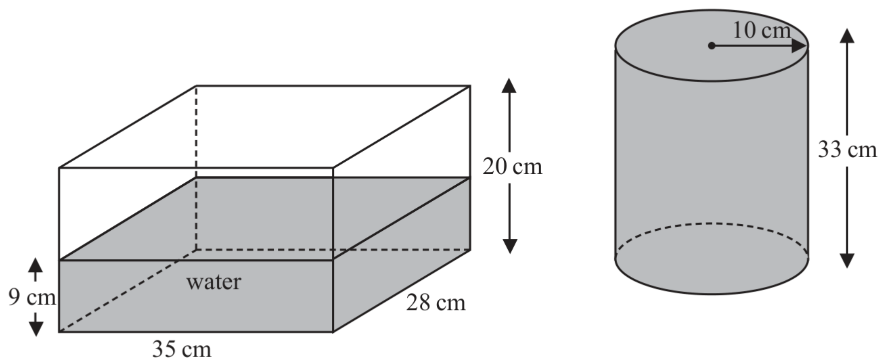
Q#: 10 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1HR - Jun 2025 | P1HR

10 The diagram shows two water containers.
One is a cuboid and one is a cylinder.

Diagram **NOT** accurately drawn



The cuboid measures 35 cm by 28 cm by 20 cm
The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm
The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION Q#: 10 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Volume & Surface Area. The tag is correct.

Solution

Capacity of the cuboid:

$$35 \times 28 \times 20 = 19600 \text{ cm}^3$$

Water already in the cuboid:

$$35 \times 28 \times 9 = 8820 \text{ cm}^3$$

Volume of the full cylinder:

$$\pi(10)^2(33) = 10367.255\ldots \text{ cm}^3$$

Total water after pouring:

$$8820 + 10367.255\ldots = 19187.255\ldots$$

Since

$$19187.255\ldots < 19600$$

the cuboid will not be completely full.

FINAL ANSWER

The cuboid will not be completely full.

PAST PAPER QUESTION

Q#: 11 Marks: 5

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1HR - Jun 2025 | P1HR

11 Zhou invests some money for 2 years.
He invests \$2500 with Bank A and \$3000 with Bank B.

Bank A

Invests \$2500

amount of money invested : total amount of interest after 2 years = 20 : 3

Bank B

Invests \$3000

4% per year compound interest for 2 years

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

\$ ....

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct, with a ratio used for Bank A.

Method

For Bank A,

$$\text{amount invested} : \text{interest} = 20 : 3$$

So the interest from Bank A is

$$2500 \times \frac{3}{20} = 375$$

For Bank B, the compound interest after 2 years is

$$3000(1.04)^2 - 3000$$

$$= 3000(1.0816) - 3000 = 244.80$$

Difference:

$$375 - 244.80 = 130.20$$

FINAL ANSWER

\$130.20

PAST PAPER QUESTION

Q#: 12 **Marks: 6**

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

12 The table gives information about the ages of 80 people in a cinema.

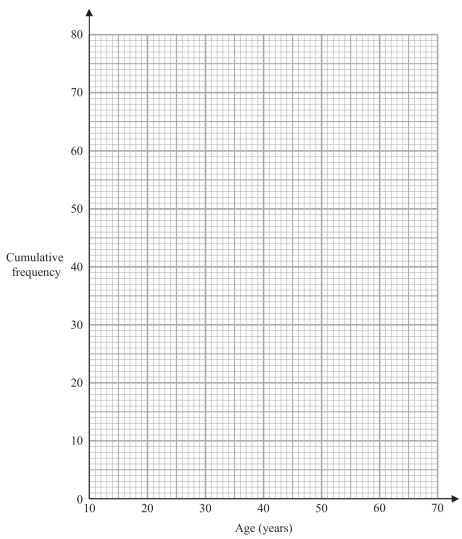
Age (n years)	**Frequency**
$10 < n \leqslant 20$	12
$20 < n \leqslant 30$	15
$30 < n \leqslant 40$	20
$40 < n \leqslant 50$	18
$50 < n \leqslant 60$	9
$60 < n \leqslant 70$	6

(a) Complete the cumulative frequency table.

Age (n years)	**Cumulative frequency**
$10 < n \leqslant 20$	
$10 < n \leqslant 30$	
$10 < n \leqslant 40$	
$10 < n \leqslant 50$	
$10 < n \leqslant 60$	
$10 < n \leqslant 70$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the percentage of the 80 people who are more than 46 years of age.
Give your answer correct to the nearest whole number.

..............................%

(3)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

The cumulative frequencies are

$$12,\ 27,\ 47,\ 65,\ 74,\ 80$$

Plot

$$(10,0),(20,12),(30,27),(40,47),(50,65),(60,74),(70,80)$$

At 46 years,

$$F(46) \approx 47 + \frac{6}{10}(65-47) = 57.8$$

So the number older than 46 is

$$80 - 57.8 = 22.2$$

$$\frac{22.2}{80} \times 100 \approx 27.75\%$$

FINAL ANSWER

about 28%.

PAST PAPER QUESTION

Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

13 Here are the numbers of eggs laid by some hens on each of 11 days in June.

9 10 12 15 17 18 19 19 20 25 26

Work out the interquartile range of the numbers of eggs.

....

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

Put the numbers in order:

$$9,\ 10,\ 12,\ 15,\ 17,\ 18,\ 19,\ 19,\ 20,\ 25,\ 26$$

$$Q_1 = 12, \qquad Q_3 = 20$$

$$\text{IQR} = 20 - 12 = 8$$

FINAL ANSWER

8.

PAST PAPER QUESTION

Q#: 14 **Marks: 2**

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

14 Work out $6.7 \times 10^{135} + 3 \times 10^{134}$

Give your answer in standard form.

....

(Total for Question 14 is 2 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Powers, roots and standard form. The tag is correct.

Method

$$6.7 \times 10^{135} + 3 \times 10^{134}$$

Write both terms using 10^{135}:

$$3 \times 10^{134} = 0.3 \times 10^{135}$$

So

$$6.7 \times 10^{135} + 0.3 \times 10^{135} = 7.0 \times 10^{135}$$

$$= 7 \times 10^{135}$$

FINAL ANSWER
7×10^{135}

PAST PAPER QUESTION

Q#: 15 Marks: 6

TOPIC: **Solving Linear Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

15 (a) Solve $\dfrac{5a+8}{3} - \dfrac{2a+5}{4} = 23$

Show clear algebraic working.

$a =$ ....

(4)

(b) Express $\left(\dfrac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

....

(2)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 6

TOPIC: **Solving Linear Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Solving Linear Equations. The equation is the main part of the question; the index form is secondary.

Method

For part (a),

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

Multiply by 12:

$$4(5a+8) - 3(2a+5) = 276$$

$$20a + 32 - 6a - 15 = 276$$

$$14a + 17 = 276$$

$$14a = 259$$

$$a = \frac{259}{14} = \frac{37}{2}$$

For part (b),

$$\left(\frac{\sqrt{y}}{3}\right)^{-1} = \frac{3}{\sqrt{y}}$$

$$= 3y^{-1/2}$$

FINAL ANSWER

(a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

PAST PAPER QUESTION

Q#: 16 **Marks: 2**

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

16 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \dfrac{101}{165}$

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is correct.

Solution

Let

$$x = 0.6121212\ldots$$

Then

$$10x = 6.121212\ldots$$

and

$$1000x = 612.121212\ldots$$

Subtract:

$$1000x - 10x = 612.121212\ldots - 6.121212\ldots$$

$$990x = 606$$

$$x = \frac{606}{990} = \frac{101}{165}$$

FINAL ANSWER

$0.6121212\ldots = \dfrac{101}{165}$.

PAST PAPER QUESTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

17 Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is correct.

Solution

Let the three consecutive multiples of 4 be

$$4n, \quad 4n+4, \quad 4n+8$$

The difference between the square of the largest and the square of the smallest is

$$(4n+8)^2 - (4n)^2$$

$$= 16n^2 + 64n + 64 - 16n^2$$

$$= 64n + 64 = 64(n+1)$$

This is always a multiple of 64.

FINAL ANSWER

The difference is always a multiple of 64.

PAST PAPER QUESTION Q#: 18 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

18 F is inversely proportional to the cube of r

$F = 6$ when $r = 2$

(a) Find a formula for F in terms of r

....

(3)

(b) Find the value of r when $F = 3072$

$r =$ ....

(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Direct and Inverse Proportion. The tag is correct.

Method

F is inversely proportional to r^3, so

$$F = \frac{k}{r^3}$$

Use $F = 6$ when $r = 2$:

$$6 = \frac{k}{2^3}$$

$$k = 48$$

So

$$F = \frac{48}{r^3}$$

When $F = 3072$,

$$3072 = \frac{48}{r^3}$$

$$r^3 = \frac{48}{3072} = \frac{1}{64}$$

$$r = \frac{1}{4}$$

FINAL ANSWER

$F = \frac{48}{r^3}$, and $r = \frac{1}{4}$.

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

19 Kannika has 9 counters.
There is a number on each counter.

1 1 2 3 3 3 5 6 6

Kannika puts the 9 counters in a bag.
She takes at random a counter from the bag and does not replace the counter.
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

....

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

The counters are

$$1,\ 1,\ 2,\ 3,\ 3,\ 3,\ 5,\ 6,\ 6$$

There are

$$\binom{9}{2} = 36$$

ways to choose two counters.
For a sum less than 5, the pairs are:

$$(1,1), \quad (1,2), \quad (1,3)$$

Counting the repeated counters gives

$$1 + 2 + 6 = 9$$

So

$$P(\text{sum} < 5) = \frac{9}{36} = \frac{1}{4}$$

FINAL ANSWER

$\frac{1}{4}$.

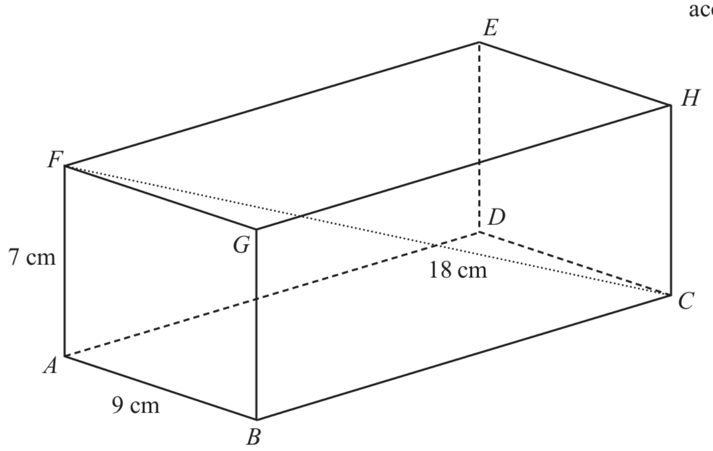
PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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Topic check. Correct.

Method

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

FINAL ANSWER

13.9 cm to 3 significant figures

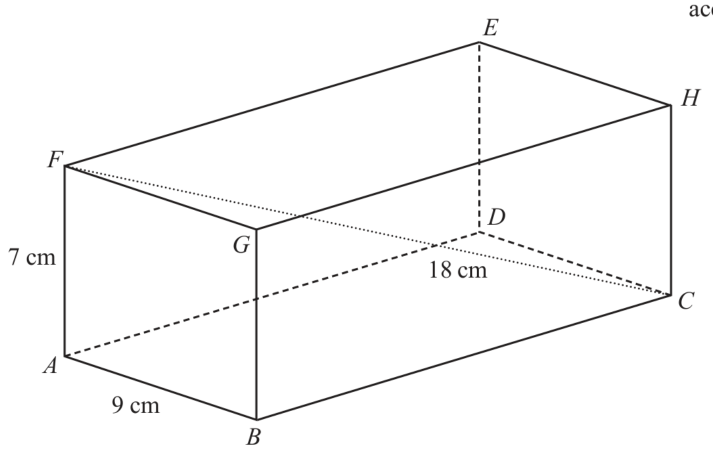
PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $AF = 7\,\text{cm}$ $FC = 18\,\text{cm}$

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

FINAL ANSWER

13.9 cm to 3 significant figures

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

21 The curve with equation $y = \text{f}(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

(a) Write down the coordinates of the turning point on the curve with equation $y = \text{f}(3x)$

(.................... , ...)

(1)

The curve **C** with equation $y = \text{g}(x)$ is transformed to give the curve **S** with equation $y = \text{g}(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

(b) Write down the value of a and the value of b

a = ...

b = ...

(2)

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(-6, 9)$.
For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(-6, 9) \to (-2, 9)$$

For part (b), a point $(4, -5)$ on $y = g(x)$ maps to $(1, -16)$ on

$$y = g(x + a) + b$$

For the x-coordinate:

$$1 + a = 4$$

$$a = 3$$

For the y-coordinate:

$$-5 + b = -16$$

$$b = -11$$

FINAL ANSWER

$(-2, 9)$, $a = 3,\ b = -11$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

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PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$(-2, 9)$, $a = 3,\ b = -11$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$
$$2x + y = 3$$

Show clear algebraic working.

....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

So

$$y = 3 - 2x$$

Substitute:

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x - 32 = 0$$

$$(5x + 8)(x - 4) = 0$$

So

$$x = -\frac{8}{5} \quad \text{or} \quad x = 4$$

Find y:

$$x = -\frac{8}{5} \Rightarrow y = \frac{31}{5}$$

$$x = 4 \Rightarrow y = -5$$

FINAL ANSWER

$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right)$ or $(4, -5)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

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$$2x + y = 3$$

Show clear algebraic working.

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(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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Substitute:

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FINAL ANSWER

$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right)$ or $(4, -5)$

PAST PAPER QUESTION Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

23

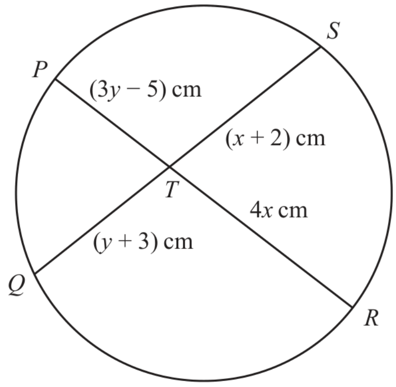

Diagram **NOT** accurately drawn

PTR and QTS are chords of a circle.

$PT = (3y - 5)$ cm $\quad QT = (y + 3)$ cm $\quad RT = 4x$ cm $\quad ST = (x + 2)$ cm

Find an expression for y in terms of x

$y =$ ....

Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For two intersecting chords in a circle,

$$PT \times TR = QT \times TS$$

Substitute the given lengths:

$$(3y - 5)(4x) = (y + 3)(x + 2)$$

Expand:

$$12xy - 20x = xy + 2y + 3x + 6$$

Collect the y terms on one side:

$$12xy - xy - 2y = 20x + 3x + 6$$

$$11xy - 2y = 23x + 6$$

Factorise:

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

FINAL ANSWER

$$\boxed{y = \frac{23x + 6}{11x - 2}}$$

PAST PAPER QUESTION **Q#: 23** **Marks: 5**

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

23

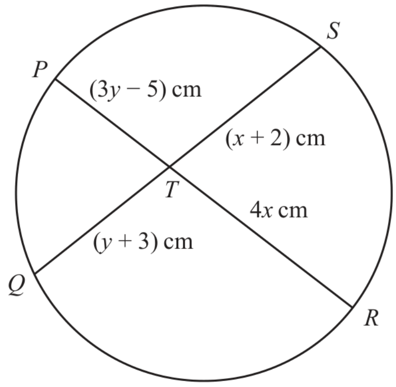

Diagram **NOT** accurately drawn

PTR and *QTS* are chords of a circle.

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Find an expression for y in terms of x

$y =$ ....

Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For two intersecting chords in a circle,

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Factorise:

$$y(11x-2) = 23x + 6$$

$$y = \frac{23x+6}{11x-2}$$

FINAL ANSWER

$$\boxed{y = \frac{23x+6}{11x-2}}$$

PAST PAPER QUESTION

Q#: 24 **Marks: 6**

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \qquad (3y - 4) \qquad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first 3 terms are

$$2x+5, \quad 3y-4, \quad 4x-2$$

Equal differences give

$$(3y-4)-(2x+5)=(4x-2)-(3y-4)$$

$$6y-6x=11$$

For an arithmetic series,

$$S_9=9(a+4d)$$

Here

$$a=2x+5$$

$$d=(3y-4)-(2x+5)=3y-2x-9$$

Given $S_9=216$:

$$9[(2x+5)+4(3y-2x-9)]=216$$

$$12y-6x=55$$

Now solve

$$6y-6x=11$$

$$12y-6x=55$$

Subtracting gives

$$6y=44$$

$$y=\frac{22}{3}$$

Then

$$6\left(\frac{22}{3}\right)-6x=11$$

$$x = \frac{11}{2}$$

FINAL ANSWER

$x = \frac{11}{2},\ y = \frac{22}{3}$

PAST PAPER QUESTION

Q#: 24 **Marks: 6**

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

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where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The first 3 terms are

$$2x + 5, \quad 3y - 4, \quad 4x - 2$$

Equal differences give

$$(3y - 4) - (2x + 5) = (4x - 2) - (3y - 4)$$

$$6y - 6x = 11$$

For an arithmetic series,

$$S_9 = 9(a + 4d)$$

Here

$$a = 2x + 5$$

$$d = (3y - 4) - (2x + 5) = 3y - 2x - 9$$

Given $S_9 = 216$:

$$9[(2x + 5) + 4(3y - 2x - 9)] = 216$$

$$12y - 6x = 55$$

Now solve

$$6y - 6x = 11$$

$$12y - 6x = 55$$

Subtracting gives

$$6y = 44$$

$$y = \frac{22}{3}$$

Then

$$6\left(\frac{22}{3}\right) - 6x = 11$$

$$x = \frac{11}{2}$$

FINAL ANSWER

$x = \frac{11}{2},\ y = \frac{22}{3}$

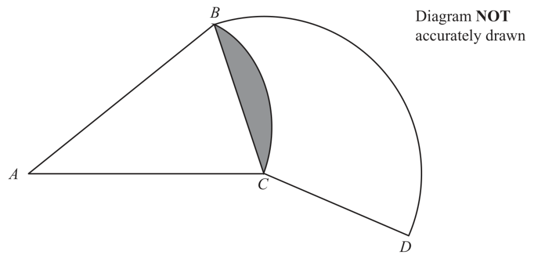
PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

25



BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Let the radius of sector BAC be R.
The shaded segment is the segment cut off by chord BC in the sector with centre A.
Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

FINAL ANSWER

$\boxed{49.4 \text{ cm}}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

25

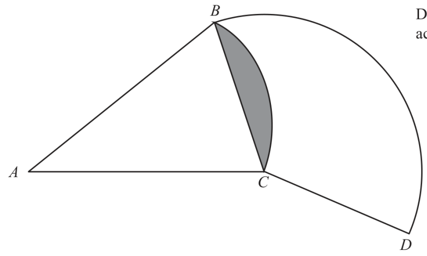


Diagram **NOT** accurately drawn

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Angle *BCD* = 130°
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Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

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PAST PAPER QUESTION

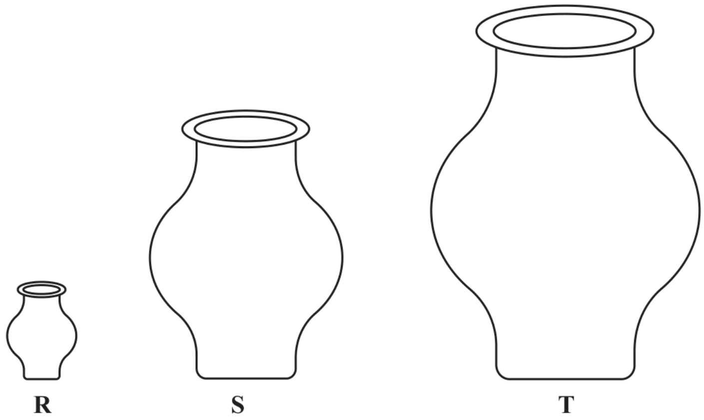
Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

26 **R**, **S** and **T** are three similar vases.

Diagram **NOT** accurately drawn



The volume of vase **S** is 72.8% more than the volume of vase **R**

The height of vase **R** is h cm
The height of vase **T** is $6h$ cm

The surface area of vase **S** is A cm^2
The surface area of vase **T** is kA cm^2

Work out the value of k

k = ...

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

Topic: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Percentages to Area & Volume of Similar Shapes, because the solution uses volume and surface area scale factors for similar shapes.

Method

The volume of vase S is 72.8% more than vase R, so

$$\frac{V_S}{V_R} = 1.728$$

For similar solids, volume scale factor is the cube of the linear scale factor:

linear scale factor from R to $S = \sqrt[3]{1.728} = 1.2$

The height of vase T is $6h$, so the linear scale factor from R to T is 6.

Therefore the linear scale factor from S to T is

$$\frac{6}{1.2} = 5$$

Surface area scale factor is the square of the linear scale factor:

$k = 5^2 = 25$

FINAL ANSWER

$k = 25$

PAST PAPER QUESTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

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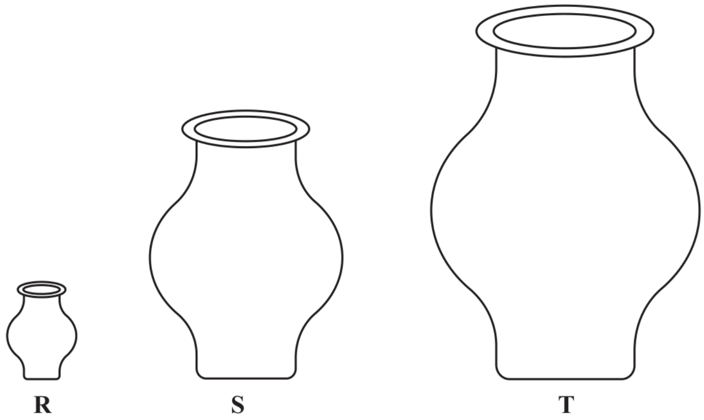

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(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

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